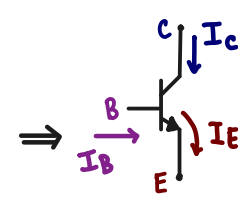
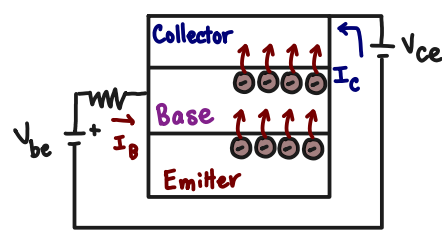


Review Bipolar Transistor



$$I_B + I_C = I_E$$

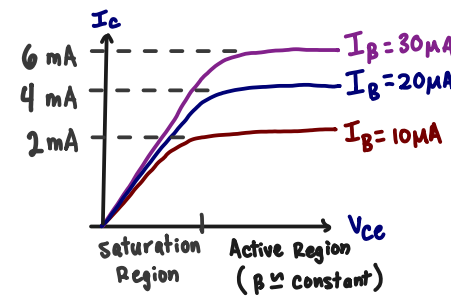
$$I_B \ll I_C + I_E$$

In the active region:

$$I = \beta I_B$$

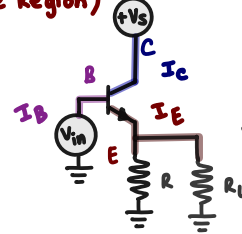
$$\beta \gg 1$$

and independent of V_{CE}



What is β ? $\beta = \frac{I_C}{I_B} = \frac{2 \text{ mA}}{10 \mu\text{A}} = \frac{2 \times 10^{-3}}{10 \times 10^{-6}} = 200$

Emitter Follower:
(Active Region)



then $V_{BE} = 0.6 \text{ V}$

$$V_{in} = V_{BE} + V_{out}$$

$$V_{out} = V_{in} - 0.6$$

Assume $V_{in} > 0.6 \text{ V}$

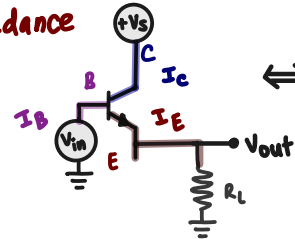
and $\frac{V_{in}}{R} \gg \beta I_B$

max possible current through R

expected collector current = I_R

$\Rightarrow$ active region: assume β

Input Impedance



$$\frac{V_B}{I_B} = Z_{in} = \frac{V_{in}}{I_{in}} = \text{Input Impedance}$$

$$\frac{\Delta V_B}{\Delta I_B} = \frac{dV_B}{dI_B} = Z_{in} \quad V_B = Z_{in} I_B$$

$$\hookrightarrow V_B = V_E - 0.6 \text{ V}, \quad I_R = I_E = \beta I_B$$

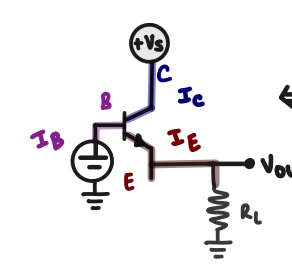
$$\text{so: } V_E = I_E R_L = V_{out}$$

$$V_B = \beta R_L I_B - 0.6 \text{ V}$$

$$Z_{in} = \frac{dV_B}{dI_B} = \beta R_L$$

* Z_{in} is β times bigger

Output Impedance



$$\hookrightarrow \frac{V_{in} - V_E}{I_E} = Z_{out} = \frac{V_{in} - V_{out}}{I_L}$$

$$\Rightarrow V_E = -I_E Z_{out} + V_{in}$$

$$\frac{dV_E}{dI_E} = -Z_{out} \rightarrow Z_{out} = -\frac{dV_E}{dI_E}$$

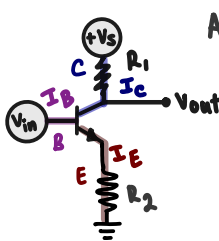
$$* V_E = V_B - 0.6 \text{ V} \quad V_{in} = I_B R_B + V_B$$

$$\hookrightarrow V_E = V_{in} - I_B R_B - 0.6 \text{ V} \quad * I_B = \frac{I_E}{1 + \beta}$$

$$V_E = V_{in} - \frac{I_E}{1 + \beta} R_B - 0.6 \text{ V} \quad Z_{out} = -\frac{dV_E}{dI_E} = \frac{R_B}{1 + \beta}$$

* Transistor lowers Z_{out} by $\beta \rightarrow$ Less voltage division @ output

Transistor Amplifier



Assume Transistor (V_{CE} big)
Is in Active Region

$\beta \gg 1$

$$I_C \approx I_B \approx I_E$$

$$V_E = V_{in} - 0.6 \text{ V}$$

$$* V_S - V_{out} = I_C R_1$$

$$\Rightarrow V_{out} = V_S - I_C R_1$$

$$V_E - (-V_S) = I_E R_2$$

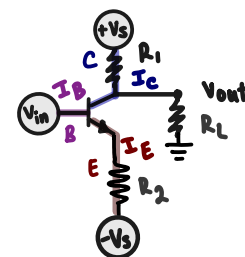
$$\hookrightarrow (V_{in} - 0.6 \text{ V}) + V_S = I_C R_2$$

$$* I_C = \frac{V_{in} - 0.6 \text{ V} + V_S}{R_2}$$

$$V_{out} = \left(-\frac{R_1}{R_2}\right) V_{in} + 0.6 \text{ V} \left(\frac{R_1}{R_2}\right) + V_S \left(1 - \frac{R_1}{R_2}\right)$$

$$V_{out} = G V_{in} + \text{constant}$$

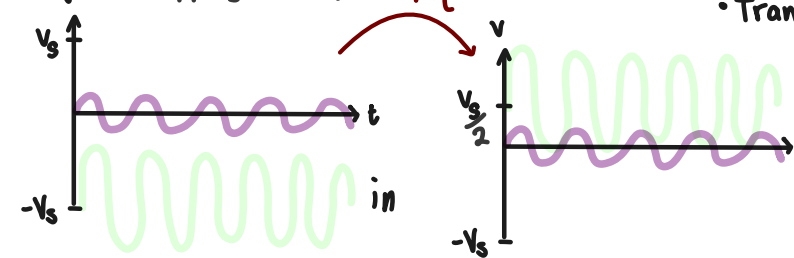
$$\text{Gain } (G) = -\frac{R_1}{R_2}$$



$$V_S - V_{out} = (I_C + I_L) R_1$$

$$= \left(\frac{V_E - V_S}{R_2} + I_L\right) R_1$$

Clipping: $R_1 > R_2 + R_L$



some constant off

Assume I_B small

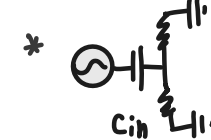
$$V_B = \frac{R_1}{R_3 + R_4} V_S$$

set V_{out} to $\frac{V_S}{2}$

by choosing ratio of $\frac{R_3}{R_4}$

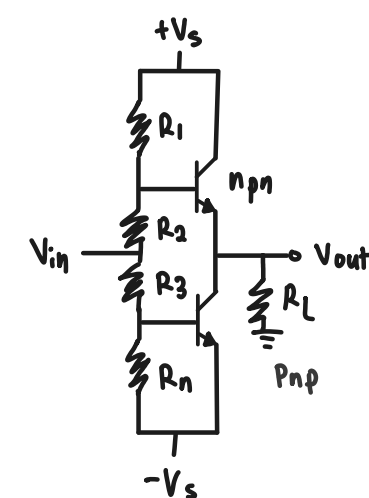
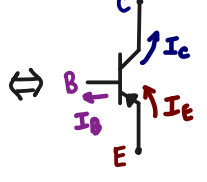
$$\text{amplification} = \frac{R_1}{R_2} = \frac{R_C}{R_E} = \frac{V_{out}}{V_{in}}$$

Assume V_{in} is an AC signal and $\omega \gg \frac{1}{R_3 || R_4 C_{in}}$

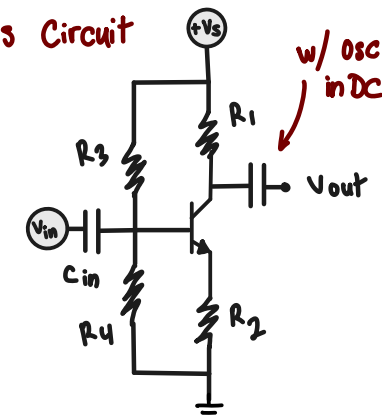


pnp Transistors

p	Collector
n	Base
p	Emitter



Bias Circuit



w/ osc. in DC

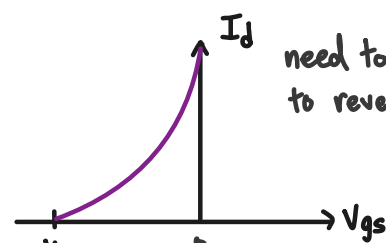
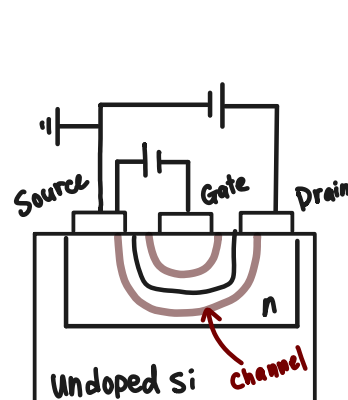
Field Effect Transistor

Junction FET (JFET)

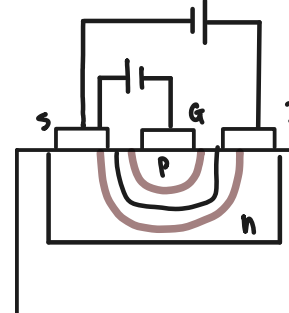
3 terminal device

Voltage on gate controls flow of current from drain to source

Used for switching + amplification



need to keep V_{GS} negative to reverse bias pn junction



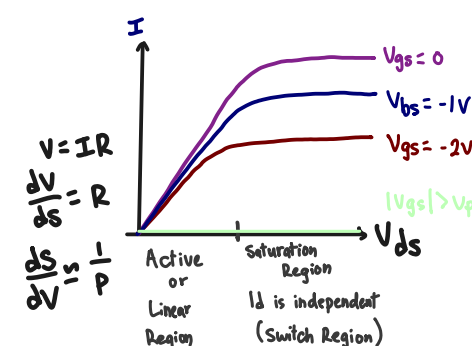
Increasing V_{BS}

Increasing R_{BS}

$$\Delta V = |V_{GS}| + V_{BS}$$

junction bias

$$V_{BS} \uparrow \Rightarrow R_{BS} \uparrow \Rightarrow I_D \approx \text{constant as } V_{BS} \text{ changes}$$



$$V = IR$$

$$\frac{dV}{dS} = R$$

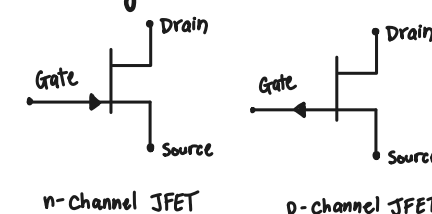
$$\frac{dS}{dV} = \frac{1}{R}$$

Active or Linear Region

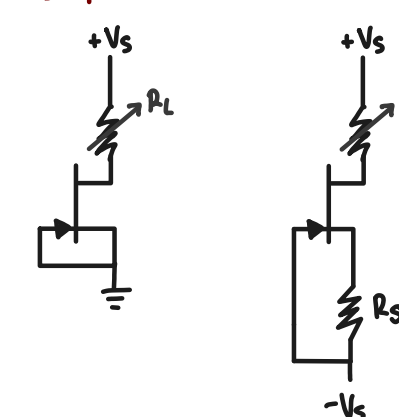
Saturation Region

I_D is independent (Switch Region)

Circuit Symbol



Simple JFET Current Source



Gate draws no current I_D